Exposé II. Topologies and Sheaves

Opening; Sections 1–2.7

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After the definition of topologies and pretopologies (no. 1) and of sheaves of sets (no. 2), no. 3 turns to the central theorem of this exposé (3.4): the existence of the sheaf associated to a presheaf. In order to cover all cases encountered in practice, the existence of this functor is proved for presheaves on a $\mathcal{U}$-site (3.0.2). Nos. 4 and 5 draw the consequences of this theorem for the behavior of inductive and projective limits in categories of sheaves. In no. 6 one defines and studies sheaves with values in arbitrary categories, and then immediately turns attention to sheaves of rings, groups, modules, etc.

1 Topologies, covering families, pretopologies

Definition 1.1. A *topology* on a category C consists, for each object X of C, of a set $J(X)$ of sieves on X, subject to the following axioms:

T 1) *Stability under base change.* For every object X of C, every sieve $R \in J(X)$, and every morphism $f : Y \to X$ with $Y \in \mathrm{Ob}(C)$, the sieve $R \times_X Y$ on Y belongs to $J(Y)$.

T 2) *Local character.* If R and R' are two sieves on X, if $R \in J(X)$, and if for every $Y \in \mathrm{Ob}(C)$ and every morphism $Y \to R$ the sieve $R' \times_X Y$ belongs to $J(Y)$, then R' belongs to $J(X)$.

T 3) For every object X of C, the maximal sieve X belongs to $J(X)$.

1.1.1

The sieves belonging to $J(X)$ will be called the *sieves covering* X, or also the *refinements of* X. From axioms (T1), (T2), and (T3), one immediately deduces that the set of sieves covering X is stable under finite intersections, and that every sieve containing a covering sieve is again a covering sieve. Thus $J(X)$, ordered by inclusion, is cofiltered (I 8).

1.1.2

Let C be a category, and let T and T' be two topologies on C. The topology T is said to be *finer than* the topology T' if, for every object X of C, every refinement of X for the topology T' is a refinement of X for the topology T. In this way one defines an order structure on the set of topologies.

1.1.3

Let $(T_i)_{i\in I}$ be a family of topologies on C. For each object X of C, the sets of sieves on X which are covering for all the topologies T_i satisfy axioms (T1), (T2), and (T3), and therefore define a topology: the *intersection topology* of the T_i, i.e. the greatest lower bound of the T_i. It is the finest of the topologies which is less fine than all the topologies T_i. It follows that the family $(T_i)_{i\in I}$ has a *least upper bound*: the intersection topology of the topologies finer than each of the T_i.

1.1.4

In particular, the assignment $J(X) =$ the set of all sieves on X is a topology finer than every topology on C; we shall call it the *discrete topology.*

There is a topology less fine than every topology on C: the topology for which $J(X) = \{X\}$ for every X of C. This topology is called the *coarse* or *chaotic* topology.

1.1.5

A category C endowed with a topology is called a *site*. The category C is called the *underlying category* of the site.

Definition 1.2. Let C be a site and let X be an object of C. A family of morphisms $(f_\alpha : X_\alpha \to X)$, $\alpha \in A$, is called *covering* if the sieve generated by the family f_α (I 4.3.3) is a sieve covering X.

1.1.6

Let C be a category. Suppose that for every object X of C we are given a set of families of morphisms with target X. Then there exists a topology T which is the least fine among the topologies for which the given families are covering, namely the intersection (1.1.3) of all topologies in question. We call this topology the *topology generated by the given sets of families of morphisms.* In general it is awkward to describe all covering sieves of this topology. The situation is more agreeable in the following case.

Definition 1.3. Let C be a category. A *pretopology* on C consists, for each object X of C, of a set $\mathrm{Cov}(X)$ of families of morphisms with target X, subject to the following axioms:

PT 0)

For every object X of C, the morphisms occurring in the families of morphisms of $\mathrm{Cov}(X)$ are base-changeable. Recall that a morphism $Y \to X$ is called base-changeable if for every morphism $Z \to X$, the fiber product $Z \times_X Y$ is representable.

PT 1)

For every object X of C, every family $(X_\alpha \to X)_{\alpha\in A}$ belonging to $\mathrm{Cov}(X)$, and every morphism $Y \to X$ with $Y \in \mathrm{Ob}(C)$, the family

$$(X_\alpha \times_X Y \to Y)_{\alpha\in A}$$

belongs to $\mathrm{Cov}(Y)$. This is stability under base change.

PT 2)

If $(X_\alpha \to X)_{\alpha \in A}$ belongs to $\mathrm{Cov}(X)$, and if for each $\alpha \in A$ the family

$$(X_{\beta_\alpha} \to X_\alpha)_{\beta_\alpha \in B_\alpha}$$

belongs to $\mathrm{Cov}(X_\alpha)$, then the family $(X_\gamma \to X)_{\gamma \in \coprod_{\alpha \in A} B_\alpha}$, where for $\gamma = (\alpha, \beta_\alpha)$ the morphism $X_\gamma \to X$ is the composite

$$X_\gamma = X_{\beta_\alpha} \to X_\alpha \to X,$$

belongs to $\mathrm{Cov}(X)$. This is stability under composition.

PT 3)

The family $X \xrightarrow{\mathrm{id}_X} X$ belongs to $\mathrm{Cov}(X)$.

1.3.1

Definition 1.1.6 allows us, for a given *pretopology* on C, to consider the *topology* on C generated by this pretopology. Note that if C is a category in which fiber products are representable, then every topology T on C can be defined by a pretopology, namely the one for which $\mathrm{Cov}(X)$ consists of *all* families covering X for the topology T.

Proposition 1.4. Let C be a category, let E be a pretopology on C, let T be the topology defined by the pretopology E (1.1.6), and let X be an object of C. Denote by $J_E(X)$ the set of sieves on X generated by the families of the pretopology, and by $J_T(X)$ the set of refinements of X for the topology T. Then $J_E(X)$ is cofinal in $J_T(X)$. In other words, for a sieve R on X to belong to $J_T(X)$, it is necessary and sufficient that there exist a sieve $R' \in J_E(X)$ such that $R' \subset R$.

Proof. For each object X of C, let $J'(X)$ be the set of sieves on X which contain a sieve of $J_E(X)$. Evidently $J'(X) \subset J_T(X)$. To show that $J'(X) = J_T(X)$, it is enough to show that the data of the $J'(X)$ define a topology on C. Now the $J'(X)$ plainly satisfy axioms (T1) and (T3). It remains to verify (T2). For this it is enough to prove that, if R' is a subsieve of $R \in J_E(X)$ such that for every $Y \to R$ the sieve $R' \times_X Y$ on Y belongs to $J'(Y)$, then R' belongs to $J'(X)$. The sieve R is generated by a family $(X_\alpha \to X)$ belonging to $\mathrm{Cov}(X)$, and for every α the sieve $R' \times_X X_\alpha$ contains a sieve generated by a family $(X_{\beta_\alpha} \to X_\alpha)$ belonging to $\mathrm{Cov}(X_\alpha)$. It follows that R' contains a sieve generated by the family $(X_{\beta_\alpha} \to X)$. Hence, by axiom (PT2), R' contains a sieve of $J_E(X)$, and therefore belongs to $J'(X)$. □

2 Sheaves of sets

Definition 2.1. Let C be a site whose underlying category is a $\mathcal{U}$-category. A presheaf F with values in $\mathcal{U}$-**Set** is called *separated* (respectively, a *sheaf*) if, for every sieve R covering an object X of C, the map

$$\mathrm{Hom}_{\widehat{C}}(X, F) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R, F)$$

is injective (respectively, bijective). The full subcategory of $\widehat{C}$ whose objects are the sheaves is called the category of sheaves of sets on C, and is most often denoted $\widetilde{C}$. When no ambiguity can result, we shall simply say the category of sheaves on C; conversely, we shall sometimes be more precise and say the category of sheaves with values in $\mathcal{U}$-**Set**, or the category of $\mathcal{U}$-sheaves.

Proposition 2.2. Let C be a $\mathcal{U}$-category, and let $\mathcal{F} = (F_i)_{i \in I}$ be a family of $\mathcal{U}$-presheaves on C. For each object X of C, denote by $J_{\mathcal{F}}(X)$ the set of sieves $R \to X$ such that, for every morphism $Y \to X$ of C with target X, the sieve $R \times_X Y$ has the following property: the map

$$\mathrm{Hom}_{\widehat{C}}(Y, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R \times_X Y, F_i)$$

is bijective (respectively, injective) for every $i \in I$. Then the sets $J_{\mathcal{F}}(X)$ define a topology on C, which is the finest topology for which each of the F_i is a sheaf (respectively, a separated presheaf).

Proof. The $J_{\mathcal{F}}(X)$ plainly satisfy axioms (T1) and (T3). It remains to show that they satisfy (T2). For this it is enough to show that:

1) if $R' \hookrightarrow R \hookrightarrow X$ are two sieves on X, with $R \in J_{\mathcal{F}}(X)$, and such that for every $Y \to R$, where Y is an object of C, the sieve $R' \times_X Y$ belongs to $J_{\mathcal{F}}(Y)$, then R' belongs to $J_{\mathcal{F}}(X)$;

2) if $R' \hookrightarrow R \hookrightarrow X$ are two sieves on X such that $R' \in J_{\mathcal{F}}(X)$, then R belongs to $J_{\mathcal{F}}(X)$.

In case 2), note that for every $Y \to R$, where Y is an object of C, the sieve $R' \times_X Y$ belongs to $J_{\mathcal{F}}(Y)$. In cases 1) and 2), the sieve R is the inductive limit of the objects Y of C lying over R (I 3.4). Inductive limits in $\widehat{C}$ are universal (I 3.3). Consequently, in cases 1) and 2), R' is the inductive limit of the $R' \times_X Y$. But in cases 1) and 2), $R' \times_X Y$ belongs to $J_{\mathcal{F}}(Y)$. Passing to the inductive limit over the objects Y of C lying over R, one deduces that the map

$$\mathrm{Hom}_{\widehat{C}}(R, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R', F_i)$$

is bijective (respectively, injective) for every $i \in I$. It follows that the maps

$$\mathrm{Hom}_{\widehat{C}}(X, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R', F_i), \qquad \mathrm{Hom}_{\widehat{C}}(X, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R, F_i)$$

are, in cases 1) and 2), bijections (respectively, injections). Moreover, hypotheses 1) and 2) are visibly stable under arbitrary base change $Y \to X$, with Y an object of C. Hence, for every object Y of C lying over X, and every $i \in I$, the maps

$$\mathrm{Hom}_{\widehat{C}}(Y, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R \times_X Y, F_i),$$
$$\mathrm{Hom}_{\widehat{C}}(Y, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R' \times_X Y, F_i)$$

are in both cases bijections (respectively, injections). This shows that R' and R belong to $J_{\mathcal{F}}(X)$. □

Corollary 2.3. Let C be a $\mathcal{U}$-category, and for every object X of C let $K(X)$ be a set of sieves on X. Assume that the $K(X)$ satisfy axiom (T1) of (1.1). For a presheaf F to be a sheaf (respectively, a separated presheaf) for the topology generated (1.1.6) by the $K(X)$, it is necessary and sufficient that, for every object X of C and every sieve $R \in K(X)$, the map

$$\mathrm{Hom}_{\widehat{C}}(X, F) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R, F)$$

be bijective (respectively, injective).

Corollary 2.4. In particular, let C be a $\mathcal{U}$-category endowed with a pretopology. For a presheaf F to be a sheaf (respectively, a separated presheaf), it is necessary and sufficient that, for every object X of C and every family $(X_\alpha \to X)$ belonging to $\mathrm{Cov}(X)$, the diagram of sets

$$F(X) \longrightarrow \prod_{\alpha \in A} F(X_\alpha) \rightrightarrows \prod_{(\alpha,\beta) \in A \times A} F(X_\alpha \times_X X_\beta)$$

be exact (respectively, that the map

$$F(X) \longrightarrow \prod_{\alpha \in A} F(X_\alpha)$$

be injective).

Proof. Apply 2.3, then I 3.5 and I 2.12.

With Corollary 2.4 one recovers the definition given in [1].

Definition 2.5. Let C be a $\mathcal{U}$-category. The *canonical topology* of C is the finest topology for which the representable functors are sheaves (2.2). A sieve covering X for the canonical topology will be called a *universally strict epimorphic sieve*. A covering family for the canonical topology will be called a *universally strict epimorphic family*. When, moreover, the morphisms of the covering family are base-changeable, the family will be called *universally effective epimorphic* [2].

Remark 2.5.1. For almost all the sites that have had to be used up to now, the topology is *less fine* than the canonical topology; in other words, the representable functors on C are sheaves, i.e. the covering families of C are universally strict epimorphic. The only exception to this rule is the site $\widehat{C}$ studied in no. 5, whose topology is *finer*, and most often strictly finer, than the canonical topology.

Proposition 2.6. For a sieve $R \hookrightarrow X$ to be universally strict epimorphic, it is necessary and sufficient that, for every object $Y \to X$ of C lying over X, and for every object Z of C, the map

$$\mathrm{Hom}_C(Y, Z) \longrightarrow \varprojlim_{C/(Y \times_X R)} \mathrm{Hom}_C(\,\cdot\,, Z)$$

be a bijection.

Proof. Immediate by applying 2.2 and then I 5.3. □

Remarks 2.7.

1) Proposition 2.6 gives a characterization of universally strict epimorphic sieves of a category C which is independent of the universe in which the presheaves take their values, under the sole condition that the Hom-sets $\mathrm{Hom}(X, Y)$ of C belong to this universe. Thus it allows one to define the canonical topology for any category C, without needing to specify the universes.

2) Let C be a site whose underlying category is a $\mathcal{U}$-category, let F be a $\mathcal{U}$-presheaf, and let $\mathcal{V}$ be a universe containing $\mathcal{U}$. The underlying category of C is a $\mathcal{V}$-category, and F may be regarded as a $\mathcal{V}$-presheaf. The $\mathcal{U}$-presheaf F is a sheaf (respectively, a separated presheaf) if and only if the $\mathcal{V}$-presheaf F is a sheaf (respectively, a separated presheaf). In other words, the condition for a presheaf F to be a sheaf (respectively, a separated presheaf) does not depend, in the sense just specified, on the universe in which the presheaf F takes its values. In particular, let C be a site and let F be a $\mathcal{U}$-presheaf; one says that F is a sheaf (respectively, a separated presheaf) if there exists a universe $\mathcal{V}$ containing $\mathcal{U}$ such that the category C is a $\mathcal{V}$-category and such that F is a $\mathcal{V}$-sheaf (respectively, a separated $\mathcal{V}$-presheaf). This property does not depend on the universe $\mathcal{V}$.

Exposé II. Topologies and Sheaves

Sections 3–6.9

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3 The sheaf associated to a presheaf

Definition 3.0.1. Let C be a site. A *topologically generating family* of C, or simply a *generating family* when no confusion can result, is a set G of objects of C such that every object of C is the target of a covering family of morphisms of C (1.2) whose sources are elements of G.

Definition 3.0.2. Let $\mathcal{U}$ be a universe. A $\mathcal{U}$*-site* is a site C whose underlying category is a $\mathcal{U}$-category (Exposé I, 1.1), and which has a small topologically generating family. If C is a $\mathcal{U}$-category, a $\mathcal{U}$*-topology* on C is a topology on C making C into a $\mathcal{U}$-site. A site C is said to be $\mathcal{U}$*-small*, or, by abuse of language, *small*, if the underlying category of C is small (Exposé I, 1.0).

3.0.3

It follows immediately from the definitions that every topology finer than a $\mathcal{U}$-topology is again a $\mathcal{U}$-topology, and that a small site is a $\mathcal{U}$-site.

Proposition 3.0.4. Let C be a $\mathcal{U}$-site and let G be a small topologically generating family of C. For $X \in \mathrm{Ob}(C)$, let $J_G(X)$ denote the set of covering sieves on X generated by a family of morphisms

$$(Y_\alpha \xrightarrow{u_\alpha} X)_{\alpha\in A}, \qquad Y_\alpha \in G.$$

Then:

1) the set $J_G(X)$ is small;

2) $J_G(X)$ is cofinal in the set $J(X)$ of all covering sieves on X, ordered by inclusion;

3) for every $R \in J_G(X)$, there is a small epimorphic family $(u_\alpha : Y \to R)_{\alpha\in A}$ with $Y \in G$ (Exposé I, 10.2).

Proof. For (1), put

$$A(X) = \coprod_{Y\in G} \mathrm{Hom}(Y, X).$$

The set $A(X)$ is small (Exposé I, 0), and $\mathrm{card}\, J_G(X) \leq 2^{\mathrm{card}\, A(X)}$. Hence $J_G(X)$ is small.

For (2), let $R \in J(X)$. Put

$$A(R) = \coprod_{Y\in G} \mathrm{Hom}(Y, R),$$

and let R' be the sieve on X generated by the family

$$Y \xrightarrow{u} R \hookrightarrow X, \qquad u \in A(R).$$

We have $R' \subset R$, and it remains only to prove that R' is covering. By axiom (T2) of 1.1, it is enough to show that, for every morphism $Z \to R$ with $Z \in \mathrm{Ob}(C)$, the sieve $R' \times_X Z$ on Z is covering. But this sieve contains the sieve generated by the family of all morphisms $Y \to Z$ with $Y \in G$, which is covering by the hypothesis on G. Hence $R' \times_X Z$ is covering, again by axiom (T2).

For (3), let $R \in J_G(X)$. The family

$$(Y \xrightarrow{u} R), \qquad u \in A(R) = \coprod_{Y \in G} \mathrm{Hom}(Y, R),$$

is epimorphic by construction. For every $Y \in \mathrm{Ob}(C)$, $\mathrm{Hom}(Y, R)$ is contained in $\mathrm{Hom}(Y, X)$, which is small. Hence $A(R)$ is small. □

3.0.5

Let C be a $\mathcal{U}$-site, let $\mathcal{V}$ be a universe such that $C \in \mathcal{V}$ and $\mathcal{U} \subset \mathcal{V}$, and let G be a $\mathcal{U}$-small topologically generating family of C. The category $\widehat{C}$ of presheaves of $\mathcal{U}$-sets on C is a $\mathcal{V}$-category (Exposé I, 1.1.1). Let X be an object of C. The set $J(X)$ of covering sieves on X, ordered by inclusion, is $\mathcal{V}$-small and cofiltered (1.1.1). For every $\mathcal{U}$-presheaf F, the inductive limit

$$\varinjlim_{R \in J(X)} \mathrm{Hom}_{\widehat{C}}(R, F)$$

is therefore representable by an element of $\mathcal{V}$ (Exposé I, 2.4.1). Moreover, by 3.0.4(3), for every $R \in J_G(X)$, the set $\mathrm{Hom}_{\widehat{C}}(R, F)$ is $\mathcal{U}$-small; since $J_G(X)$ is a $\mathcal{U}$-small cofinal subset of $J(X)$ (loc. cit.), it follows from Exposé I, 2.4.2 that

$$\varinjlim_{R \in J(X)} \mathrm{Hom}_{\widehat{C}}(R, F)$$

is $\mathcal{U}$-small. We choose, for each F and X, an element of $\mathcal{U}$ representing this inductive limit, and put

$$LF(X) = \varinjlim_{R \in J(X)} \mathrm{Hom}_{\widehat{C}}(R, F).$$

If $g : Y \to X$ is a morphism of C, the base-change functor $g^* : J(X) \to J(Y)$ defines a map

$$LF(g) : LF(X) \longrightarrow LF(Y),$$

making $X \mapsto LF(X)$ into a presheaf on C.

Since the morphism $\mathrm{id}_X : X \to X$ is an element of $J(X)$, for every object X of C one has a map

$$\ell(F)(X) : F(X) \longrightarrow LF(X),$$

and these maps plainly define a morphism of presheaves

$$\ell(F) : F \longrightarrow LF.$$

It is also clear that $F \mapsto LF$ is functorial in F, and that the morphisms $\ell(F)$ define a natural transformation

$$\ell : \mathrm{Id} \longrightarrow L.$$

Finally, let $R \hookrightarrow X$ be a refinement of X. The definition of $LF(X)$ and Exposé I, 1.4 give a map

$$Z_R : \mathrm{Hom}_{\widehat{C}}(R, F) \longrightarrow \mathrm{Hom}_{\widehat{C}}(X, LF).$$

For every morphism $g : Y \to X$ of C, the definition of LF shows that the square

$$\begin{array}{ccc} \mathrm{Hom}_{\widehat{C}}(R, F) & \xrightarrow{Z_R} & \mathrm{Hom}_{\widehat{C}}(X, LF) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_{\widehat{C}}(R \times_X Y, F) & \xrightarrow{Z_{R \times_X Y}} & \mathrm{Hom}_{\widehat{C}}(Y, LF) \end{array} \qquad (*)$$

is commutative, where the vertical arrows are the evident ones.

We now copy the standard argument of Demazure.

Lemma 3.1.

1) For every refinement $i_R : R \hookrightarrow X$ and every $u : R \to F$, the diagram

$$\begin{array}{ccc} R & \xrightarrow{i_R} & X \\ {\scriptstyle u} \downarrow & & \downarrow {\scriptstyle Z_R(u)} \\ F & \xrightarrow{\ell(F)} & LF \end{array} \qquad (**)$$

is commutative.

2) For every morphism $v : X \to LF$, there exists a refinement R of X and a morphism $u : R \to F$ such that $Z_R(u) = v$.

3) Let Y be an object of C, and let $u, v : Y \rightrightarrows F$ be two morphisms such that $\ell(F) \circ u = \ell(F) \circ v$. Then the equalizer, or kernel, of the pair (u, v) is a refinement of Y.

4) Let R and R' be two refinements of X, and let $u : R \to F$ and $u' : R' \to F$ be two morphisms. Then $Z_R(u) = Z_{R'}(u')$ if and only if u and u' coincide on some refinement

$$R'' \hookrightarrow R \times_X R'.$$

Proof. Only assertion (1) is nontrivial. We must prove

$$Z_R(u) \circ i_R = \ell(F) \circ u.$$

By Exposé I, 3.4 it is enough to prove equality after composition with every morphism $g : Y \to R$, where Y is an object of C. Let $f = i_R \circ g$, and consider $R \times_X Y$. The canonical monomorphism $R \times_X Y \to Y$ has a section, hence is an isomorphism by 1.3(3). By definition of $\ell(F)$, the morphism $Z_{R\times_X Y}(u \circ g')$ is $\ell(F) \circ u \circ g$. On the other hand, the commutativity of $(*)$ gives

$$Z_{R\times_X Y}(u \circ g') = Z_R(u) \circ f.$$

Thus $\ell(F) \circ u \circ g = Z_R(u) \circ i_R \circ g$, as required. $\square$

Proposition 3.2.

1) The functor L is left exact (Exposé I, 2.3.2).

2) For every presheaf F, LF is a separated presheaf.

3) A presheaf F is separated if and only if the morphism $\ell(F) : F \to LF$ is a monomorphism. In that case LF is a sheaf.

4) The following properties are equivalent:

   (i) $\ell(F) : F \to LF$ is an isomorphism;

   (ii) F is a sheaf.

Proof. For (1), it is enough, by Exposé I, 3.1, to prove that for every object X of C, the functor $F \mapsto LF(X)$ commutes with finite projective limits. But for every $R \hookrightarrow X$ in $J(X)$, the functor

$$F \longmapsto \mathrm{Hom}_{\widehat{C}}(R, F)$$

commutes with projective limits, by the definition of projective limit; and the filtered inductive limit $\varinjlim_{J(X)}$ commutes with finite projective limits because $J(X)$ is a cofiltered set (Exposé I, 2.7).

For (2), let X be an object of C, and let $f, g : X \rightrightarrows LF$ be two morphisms which coincide on a refinement $R \hookrightarrow X$. By Lemma 3.1(2), there is a covering sieve $R' \hookrightarrow X$, which we may suppose contained in R, and two morphisms $u, v : R' \rightrightarrows F$ such that $Z_{R'}(u) = f$ and $Z_{R'}(v) = g$. By Lemma 3.1(1), $\ell(F) \circ u = \ell(F) \circ v$. Hence, by Lemma 3.1(4), u and v coincide on a refinement $R'' \hookrightarrow R'$. If w denotes the restriction of u to R'', then

$$Z_{R''}(w) = Z_{R'}(u) = Z_{R'}(v),$$

and consequently $f = g$. Thus LF is separated.

For (3), if F is separated, then $\ell(F)$ is a monomorphism, because a filtered inductive limit of monomorphisms is a monomorphism. Conversely, if $\ell(F)$ is a monomorphism, then F is a subpresheaf of a separated presheaf, hence is separated. We now prove that LF is then a sheaf. Let $i : R \hookrightarrow X$ be a covering sieve of an object X of C, and let $u : R \to LF$ be a morphism. It is enough to prove that u factors through X. Put

$$R' = F \times_{LF} R, \qquad v = Z_{R'}(\mathrm{pr}_1).$$

To prove $u = v \circ i$, it is enough, by Exposé I, 3.4, to test after every morphism $m : Y \to R$ with Y an object of C. Put $R'' = Y \times_R R'$, with projections p_1, p_2. The projection $R'' \to Y$ is a monomorphism making R'' into a covering sieve of Y by Lemma 3.1(2). We have

$$v \circ i \circ p_2 = \ell(F) \circ \mathrm{pr}_1$$

by Lemma 3.1(1), and hence

$$v \circ i \circ p_2 = u \circ m \circ p_2.$$

Pulling back along p_1 gives

$$v \circ i \circ m \circ p_2 = u \circ m \circ p_2.$$

Since LF is separated, $v \circ i \circ m = u \circ m$. This proves the sheaf condition.

Assertion (4) is clear from (3). □

Remark 3.3. Let $J'(X)$ be a cofinal subset of $J(X)$. Then

$$LF(X) = \varinjlim_{R \in J'(X)} \operatorname{Hom}_{\widehat{C}}(R, F).$$

In particular, if the topology of C is defined by a pretopology $X \mapsto \operatorname{Cov}(X)$ (1.1.3), then the functor L can be described by means of the covering families of $\operatorname{Cov}(X)$ (Exposé I, 2.12 and 3.5). Making the formulae explicit, one recovers the usual construction.

Theorem 3.4. Let C be a $\mathcal{U}$-site. The inclusion functor

$$i : \widetilde{C} \hookrightarrow \widehat{C}$$

from sheaves to presheaves has a left adjoint

$$\underline{a} : \widehat{C} \longrightarrow \widetilde{C},$$

which is left exact (Exposé I, 2.3.2). The functor $i \circ \underline{a}$ is canonically isomorphic to $L \circ L$ (cf. 3.0.5). For every presheaf F, the adjunction morphism

$$F \longrightarrow i\, \underline{a}(F)$$

is, through the preceding isomorphism, the composite

$$F \xrightarrow{\ell(F)} LF \xrightarrow{\ell(LF)} L(LF).$$

Definition 3.5. The sheaf $\underline{a}F$ is called the *sheaf associated to the presheaf* F, or the *sheafification* of F.

Theorem 3.4 follows immediately from Proposition 3.2.

Proposition 3.6. Let C be a $\mathcal{U}$-site and let $\mathcal{V} \supset \mathcal{U}$ be a universe. Denote by

$$\widehat{C}_{\mathcal{U}}, \quad \widetilde{C}_{\mathcal{U}}$$

the categories of $\mathcal{U}$-presheaves and $\mathcal{U}$-sheaves, and by

$$\widehat{C}_{\mathcal{V}}, \quad \widetilde{C}_{\mathcal{V}}$$

the categories of $\mathcal{V}$-presheaves and $\mathcal{V}$-sheaves. Let

$$\underline{a}_{\mathcal{U}} : \widehat{C}_{\mathcal{U}} \to \widetilde{C}_{\mathcal{U}}, \qquad \underline{a}_{\mathcal{V}} : \widehat{C}_{\mathcal{V}} \to \widetilde{C}_{\mathcal{V}}$$

be the corresponding associated-sheaf functors. Then the diagram

$$\begin{array}{ccc} \widehat{C}_{\mathcal{U}} & \xrightarrow{\underline{a}_{\mathcal{U}}} & \widetilde{C}_{\mathcal{U}} \\ \downarrow & & \downarrow \\ \widehat{C}_{\mathcal{V}} & \xrightarrow{\underline{a}_{\mathcal{V}}} & \widetilde{C}_{\mathcal{V}} \end{array}$$

where the vertical functors are the canonical inclusions, commutes up to canonical isomorphism.

Proof. This follows from the construction of the functors $\underline{a}_{\mathcal{U}}$ and $\underline{a}_{\mathcal{V}}$ (3.4 and 3.0.5). $\square$

4 Exactness properties of the category of sheaves

The exactness properties of the category of sheaves are deduced from those of the category of presheaves by means of Theorem 3.4. This section makes that philosophy explicit in a number of typical statements which are among the most useful.

Theorem 4.1. Let C be a $\mathcal{U}$-site, let $\widetilde{C}$ be the category of sheaves, let

$$\underline{a} : \widehat{C} \longrightarrow \widetilde{C}$$

be the associated-sheaf functor, and let

$$i : \widetilde{C} \longrightarrow \widehat{C}$$

be the inclusion functor.

1) The functor $\underline{a}$ commutes with inductive limits and is exact.

2) $\mathcal{U}$-inductive limits in $\widetilde{C}$ are representable. For every small category I and every functor $E : I \to \widetilde{C}$, the canonical morphism

$$\varinjlim_I E \longrightarrow \underline{a}\left(\varinjlim_I iE\right)$$

is an isomorphism.

3) $\mathcal{U}$-projective limits in $\widetilde{C}$ are representable. For every object X of C, the functor $F \mapsto F(X)$ on $\widetilde{C}$ commutes with projective limits; equivalently, the inclusion $i : \widetilde{C} \to \widehat{C}$ commutes with projective limits.

Proof. These properties follow essentially from Theorem 3.4 and Exposé I, 2.11. $\square$

Thus, in the category of sheaves, products indexed by an element of $\mathcal{U}$, fiber products, sums indexed by an element of $\mathcal{U}$, amalgamated sums, kernels, cokernels, images, and coimages are representable.

Corollary 4.1.1. Let C be a $\mathcal{U}$-site and let F be a sheaf of sets on C. The canonical homomorphism

$$\varinjlim_{C/F} \underline{a}X \longrightarrow F$$

is an isomorphism.

Proof. This follows from Exposé I, 3.4 and the fact that $\underline{a}$ commutes with inductive limits. □

Proposition 4.2. Every morphism of $\widetilde{C}$ which is both an epimorphism and a monomorphism is an isomorphism.

Proof. Let $u : G \to H$ be a morphism of $\widetilde{C}$ which is both an epimorphism and a monomorphism. First note that u is a monomorphism of presheaves (4.1(1)). Form the amalgamated sum $H \amalg_G H = K$ and the two canonical morphisms $i_1, i_2 : H \rightrightarrows K$ in the category of presheaves. Since u is a monomorphism of presheaves, the square

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle i_2} \\ H & \xrightarrow{i_1} & K \end{array}$$

is immediately verified to be both cartesian and cocartesian (Exposé I, 10.1). Applying the associated-sheaf functor, we get a cartesian and cocartesian square in the category of sheaves:

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle \underline{a}i_2} \\ H & \xrightarrow{\underline{a}i_1} & \underline{a}K. \end{array} \tag{*}$$

Since u is an epimorphism of sheaves, $\underline{a}i_1$ is an isomorphism; since the square $(*)$ is cartesian, u is an isomorphism. □

Proposition 4.3.

1) Representable inductive limits in $\widetilde{C}$ are universal (Exposé I, 2.5).

2) Every epimorphic family of morphisms is an effective universal epimorphic family (2.6).

3) Every equivalence relation is effective and universal (Exposé I, 10.6).

4) $\mathcal{U}$-filtered inductive limits commute with finite projective limits (Exposé I, 2.6).

Proof. Assertions (1)–(4) hold in the category of sets and hence in the category of presheaves of sets (Exposé I, 3.1). Assertions (1) and (4) then follow at once from the corresponding assertions for presheaves and from Theorem 4.1.

We prove (3). Let $p_1, p_2 : R \rightrightarrows X$ be an equivalence relation in $\widetilde{C}$. It is also an equivalence relation of presheaves (4.1.1). Hence there exists a morphism of presheaves $u : X \to Y$ such that

$$\begin{array}{ccc} R & \xrightarrow{p_2} & X \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle u} \\ X & \xrightarrow{u} & Y \end{array}$$

is both cartesian and cocartesian in the category of presheaves. Applying the associated-sheaf functor, one obtains a cartesian and cocartesian square in the category of sheaves

(4.1(1)). Thus $R \rightrightarrows X$ is an effective equivalence relation. Since every equivalence relation is effective by the preceding argument, every equivalence relation is universally effective.

We prove (2). Let $(u_i : X_i \to X)_{i\in I}$ be an epimorphic family in $\widetilde{C}$. By 2.6 and Exposé I, 2.12, it is enough to prove:

a) for every morphism of sheaves $v : Y \to X$, the family

$$\mathrm{pr}_{2,i} : X_i \times_X Y \to Y, \qquad i \in I,$$

is epimorphic in $\widetilde{C}$;

b) for every sheaf Z, the diagram of sets

$$\mathrm{Hom}(X, Z) \longrightarrow \prod_{i\in I} \mathrm{Hom}(X_i, Z) \rightrightarrows \prod_{(i,k)\in I\times I} \mathrm{Hom}(X_i \times_X X_k, Z)$$

is exact.

Let X' be the union of the images of the morphisms u_i, in the sense of presheaves, and let $j : X' \hookrightarrow X$ be the canonical injection. For every i, the morphism $u_i : X_i \to X$ factors as $X_i \xrightarrow{u'_i} X' \xrightarrow{j} X$, and the family $u'_i : X_i \to X'$ is epimorphic in $\widehat{C}$. Since j is a monomorphism, the morphism $\underline{a}j : \underline{a}X' \to X$ is a monomorphism of sheaves (4.1). Since $u_i = \underline{a}j\, \underline{a}u'_i$ for each i, $\underline{a}j$ is also an epimorphism of sheaves. Hence $\underline{a}j$ is an isomorphism by Proposition 4.2.

Let $v : Y \to X$ be a morphism of sheaves. After base change we obtain a morphism $j_v : X' \times_X Y \to Y$ and morphisms $u'_{i,v} : X_i \times_X Y \to X' \times_X Y$. Since $\underline{a}$ commutes with fiber products, $\underline{a}j_v$ is an isomorphism. Since epimorphic families of presheaves remain epimorphic after base change, the family $u'_{i,v}$ is epimorphic in $\widehat{C}$. As $\underline{a}$ commutes with inductive limits, the family

$$\underline{a}u'_{i,v} : X_i \times_X Y \to \underline{a}(X' \times_X Y)$$

is epimorphic in $\widetilde{C}$. This proves (a).

For (b), let Z be a sheaf. Since $\underline{a}j$ is an isomorphism, the map

$$\mathrm{Hom}(X, Z) \longrightarrow \mathrm{Hom}(X', Z)$$

is a bijection. Since the u'_i form an epimorphic family in $\widehat{C}$, the diagram

$$\mathrm{Hom}(X', Z) \longrightarrow \prod_{i\in I} \mathrm{Hom}(X_i, Z) \rightrightarrows \prod_{(i,k)\in I\times I} \mathrm{Hom}(X_i \times_{X'} X_k, Z)$$

is exact. Finally, since X' is a subpresheaf of X, the presheaf $X_i \times_{X'} X_k$ is canonically isomorphic to $X_i \times_X X_k$. This gives (b). □

Remarks 4.3.1.

1) Proposition 4.3(2) formally gives a second proof of Proposition 4.2. In any category, a morphism which is both an effective universal epimorphism and a monomorphism is plainly an isomorphism.

2) From the preceding propositions it follows that every morphism in the category of sheaves factors uniquely as an effective epimorphism followed by an effective monomorphism. For non-additive categories, this property is the natural analogue of axiom (AB2) for abelian categories, namely coim $\simeq$ im [Tohoku].

4.4.0

Let C be a $\mathcal{U}$-site. The functor $h : C \to \widehat{C}$ (Exposé I, 1.3.1), composed with the associated-sheaf functor, gives a functor

$$\epsilon_C : C \longrightarrow \widetilde{C},$$

called the *canonical functor* from C to $\widetilde{C}$. It will be used constantly below. The functor ϵ_C commutes with finite projective limits. When the topology of C is less fine than the canonical topology, ϵ_C is fully faithful and commutes with projective limits; in that case it is, moreover, defined even if C need not have a small topologically generating family. We shall not study in detail the behavior of ϵ_C with respect to inductive limits (cf. SGA 3, IV). We shall, however, need the following propositions.

Theorem 4.4. Let C be a $\mathcal{U}$-site and let $(u_i : X_i \to X)_{i \in I}$ be a family of morphisms of C with target X. The following conditions are equivalent:[1]

(i) The family

$$(\epsilon_C u_i : \epsilon_C X_i \to \epsilon_C X)_{i \in I}$$

is an epimorphic family in $\widetilde{C}$ (Exposé I, 10.2).

(ii) The family $(u_i : X_i \to X)_{i \in I}$ is a covering family of C (1.2).

Proof. (ii) $\Rightarrow$ (i). Let $R \hookrightarrow X$ be the sieve generated by the u_i (Exposé I, 4.3.3), and let $u_i' : X_i \to R$ be the morphisms induced by the u_i. The family u_i' is epimorphic in $\widehat{C}$, and R is a covering sieve on X. Hence, for every sheaf F, the map

$$\mathrm{Hom}(X, F) \longrightarrow \mathrm{Hom}(R, F)$$

is bijective, while

$$\mathrm{Hom}(R, F) \longrightarrow \prod_i \mathrm{Hom}(X_i, F)$$

is injective. Thus

$$\mathrm{Hom}(X, F) \longrightarrow \prod_i \mathrm{Hom}(X_i, F)$$

is injective, and consequently

$$\mathrm{Hom}(\underline{a}X, F) \longrightarrow \prod_i \mathrm{Hom}(\underline{a}X_i, F)$$

is injective. This is exactly (i).

(i) $\Rightarrow$ (ii). With the notation just introduced, let $J_R : R \hookrightarrow X$ be the canonical inclusion of the sieve generated by the u_i. From (i) it follows that the morphism of sheaves

$$\underline{a}J_R : \underline{a}R \longrightarrow \underline{a}X$$

[1]When the site C need not have a small topologically generating family and when the topology of C is less fine than the canonical topology, one has (ii) $\Rightarrow$ (i); see the proof of 4.4.

is both an epimorphism and a monomorphism. Hence it is an isomorphism by Proposition 4.2. With the notation of Section 3, we have a commutative diagram

$$\begin{array}{ccccc} X & \xrightarrow{\ell(X)} & LX & \xrightarrow{\ell(LX)} & \underline{a}X \\ \uparrow J_R & & \uparrow LJ_R & & \uparrow \wr \\ R & \xrightarrow{\ell(R)} & LR & \xrightarrow{\ell(LR)} & \underline{a}R. \end{array}$$

By Lemma 3.1(2), there exists a covering sieve $J_{S_1} : S_1 \to X$ and a morphism $u_1 : S_1 \to LR$ such that

$$\ell(LX) \circ \ell(X) \circ J_{S_1} = \underline{a}J_R \circ \ell(LR) \circ u_1 = \ell(LX) \circ LJ_R \circ u_1. \tag{4.6.1}$$

Put $m = \ell(X) \circ J_{S_1}$ and $n = LJ_R \circ u_1$. These are morphisms $S_1 \to LX$. Let $S_2 \hookrightarrow S_1$ be the equalizer of the pair (m, n). For every object Y of C and every morphism $\alpha : Y \to S_1$, equation (4.6.1) gives

$$\ell(LX)m\alpha = \ell(LX)n\alpha.$$

By Lemma 3.1(3), the equalizer $S_2 \times_{S_1} Y$ of $m\alpha$ and $n\alpha$ is a covering sieve on Y. Hence, by axiom (T2), $J_{S_2} : S_2 \hookrightarrow X$ is a covering sieve on X. Thus we have a morphism $u_2 : S_2 \to LR$ and a commutative diagram

$$\begin{array}{ccc} S_2 & \xrightarrow{J_{S_2}} & X \\ {\scriptstyle u_2} \downarrow & & \downarrow {\scriptstyle \ell(X)} \\ LR & \xrightarrow{LJ_R} & LX \end{array} \tag{4.6.2}$$

with $R \hookrightarrow X$ inserted in the evident way through J_R.

Let Y be an object of C and let $\beta : Y \to S_2$ be a morphism. By Lemma 3.1(1), there exists a covering sieve $J_{Q_1} : Q_1 \hookrightarrow Y$ and a morphism $v_1 : Q_1 \to R$ such that

$$u_2\beta J_{Q_1} = \ell(R)v_1.$$

Put

$$f = J_{S_2}\beta J_{Q_1}, \qquad g = J_R v_1.$$

Let Q_2 be the equalizer of the pair $f, g : Q_1 \rightrightarrows X$, and let $J_{Q_2} : Q_2 \hookrightarrow Y$ be the canonical inclusion. For every object Z of C and every morphism $\gamma : Z \to Q_1$, the commutativity of (4.6.2) gives

$$\ell(X)f\gamma = LJ_R\, u_2\beta J_{Q_1}\gamma = LJ_R\, \ell(R)v_1\gamma = \ell(X)J_R v_1\gamma = \ell(X)g\gamma.$$

By Lemma 3.1(3), the equalizer $Q_2 \times_{Q_1} Z$ of $f\gamma$ and $g\gamma$ is a covering sieve on Z. By axiom (T2), $J_{Q_2} : Q_2 \hookrightarrow Y$ is a covering sieve on Y. Hence, for every morphism $\beta : Y \to S_2$, there exists a covering sieve $Q_2 \hookrightarrow Y$ and a morphism $v_2 : Q_2 \to R$ making

$$\begin{array}{ccccc} Q_2 & \xrightarrow{v_2} & & & R \\ \downarrow & & & & \downarrow J_R \\ Y & \xrightarrow{\beta} & S_2 & \xrightarrow{J_{S_2}} & X \end{array}$$

commutative.

Now let $S_2 \cap R$ denote the fiber product of S_2 and R over X. The sieve

$$(S_2 \cap R) \times_{S_2} Y$$

on Y contains Q_2; hence it is covering. Axiom (T2) implies that $S_2 \cap R$ is a covering sieve on X. Since R contains $S_2 \cap R$, R is itself a covering sieve on X. □

Corollary 4.4.4. Let T be the topology of the $\mathcal{U}$-site C. Let T' (respectively T'') be the finest topology on C among those for which the objects of $\widetilde{C}$ are sheaves (respectively separated presheaves) (2.2). Then

$$T = T' = T''.$$

Indeed, trivially $T \leq T' \leq T''$, while Theorem 4.4 and Proposition 2.2 imply $T'' \leq T$. □

Definition 4.5. An *initial object* of a category A is an object ϕ_A representing the empty inductive limit, i.e. such that for every $X \in \mathrm{Ob}(A)$ there is exactly one arrow $\phi_A \to X$. An object ϕ_A is called *strict initial* if it is initial and every morphism with target ϕ_A is an isomorphism.

Let $(S_i)_{i \in I}$ be a family of objects of a category A. Suppose that the sum $S = \coprod_{i \in I} S_i$ is representable. The sum S is said to be *disjoint* if the structural morphisms $S_i \to S$ are base-changeable monomorphisms and, for every pair i, j with $i \neq j$, the fiber product $S_i \times_S S_j$ is an initial object of A. The sum is said to be *universally disjoint* if it is disjoint and remains a disjoint sum after every base change $T \to S$. It follows that, for every $i \neq j$, the objects $S_i \times_S S_j$ are strict initial objects of A.

Example 4.5.1. In the category of sets, direct sums are disjoint and universal. Hence the same is true in every category of presheaves of sets (Exposé I, 3.1), and therefore in every category of sheaves of sets on a $\mathcal{U}$-site (4.1). In particular, the initial object of $\widetilde{C}$ is strict.

Proposition 4.6. Let C be a $\mathcal{U}$-category and let $(s_i : X_i \to X)_{i \in I}$ be a family of morphisms of C. For every $\mathcal{U}$-topology $\mathcal{T}$ on C (3.0.2), let $\widetilde{C}_{\mathcal{T}}$ be the corresponding category of sheaves, and let

$$\epsilon_{\mathcal{T}} : C \to \widetilde{C}_{\mathcal{T}}$$

be the corresponding canonical functor.

1) Let $\mathcal{T}$ be a $\mathcal{U}$-topology such that

$$\coprod_{i \in I} \epsilon_{\mathcal{T}}(X_i) \xrightarrow{(\epsilon_{\mathcal{T}}(s_i))} \epsilon_{\mathcal{T}}(X)$$

is an isomorphism. Then, for every topology $\mathcal{T}'$ finer than $\mathcal{T}$, the morphism

$$\coprod_{i \in I} \epsilon_{\mathcal{T}'}(X_i) \xrightarrow{(\epsilon_{\mathcal{T}'}(s_i))} \epsilon_{\mathcal{T}'}(X)$$

is an isomorphism.

2) Let $\mathcal{T}$ be a $\mathcal{U}$-topology on C. The following properties (i) and (ii) are equivalent:

(i) The following three conditions hold:

(a) The family $(s_i : X_i \to X)_{i \in I}$ is covering for $\mathcal{T}$.

(b) For every $i \in I$, the diagonal morphism of presheaves

$$\Delta_i : X_i \hookrightarrow X_i \times_X X_i$$

is transformed by the associated-sheaf functor for $\mathcal{T}$ into an isomorphism; this is automatically the case if the s_i are monomorphisms.

(c) For every pair $i \neq j$, the presheaf $X_i \times_X X_j$ is transformed by the associated-sheaf functor for $\mathcal{T}$ into the initial object of $\widetilde{C}_{\mathcal{T}}$.

(ii) $\epsilon_{\mathcal{T}}(X)$ is the sum of the $\epsilon_{\mathcal{T}}(X_i)$.

Proof. For (1), a sheaf for $\mathcal{T}'$ is a sheaf for $\mathcal{T}$. Hence, for every sheaf F for $\mathcal{T}'$, the morphism

$$\mathrm{Hom}_{\widehat{C}}(X, F) \longrightarrow \prod_{i \in I} \mathrm{Hom}_{\widehat{C}}(X_i, F)$$

is an isomorphism. This implies the assertion.

For (2), by definition property (ii) holds if and only if the morphism of presheaves

$$\Phi = (h(s_i))_{i \in I} : \coprod_i h(X_i) \longrightarrow h(X)$$

is transformed by the associated-sheaf functor into an isomorphism; equivalently, by Proposition 4.2, $\underline{a}(\Phi)$ is both an epimorphism and a monomorphism of sheaves. By Theorem 4.4, $\underline{a}(\Phi)$ is an epimorphism if and only if (a) holds. The diagonal

$$\Delta : \coprod_i h(X_i) \longrightarrow \left(\coprod_i h(X_i)\right) \times_{h(X)} \left(\coprod_i h(X_i)\right)$$

is the direct sum of morphisms $\Delta_{i,j}$, indexed by $I \times I$, defined as follows:

- if $i = j$, $\Delta_{i,i}$ is the diagonal

$$h(X_i) \to h(X_i) \times_{h(X)} h(X_i);$$

- if $i \neq j$, $\Delta_{i,j}$ is the morphism

$$\phi_{\widehat{C}} \to h(X_i) \times_{h(X)} h(X_j),$$

  where $\phi_{\widehat{C}}$ denotes the initial object of $\widehat{C}$.

The morphism $\underline{a}(\Phi)$ is a monomorphism if and only if $\underline{a}(\Delta)$ is an isomorphism; and by Theorem 4.1, $\underline{a}(\Delta)$ is an isomorphism if and only if each $\underline{a}(\Delta_{i,j})$ is an isomorphism, i.e. if and only if (b) and (c) hold. $\square$

Corollary 4.6.1. Let C be a $\mathcal{U}$-category and let X be an object of C.

1) Let $\mathcal{T}$ be a $\mathcal{U}$-topology on C. The object X of C is transformed by the associated-sheaf functor for $\mathcal{T}$ into the initial object of $\widetilde{C}_{\mathcal{T}}$ if and only if the empty sieve covers X.

2) When X is a strict initial object of C, the sheaf associated to X, for every $\mathcal{U}$-topology finer than the canonical topology, is an initial object.

Proof. For (1), take the indexing set I in Proposition 4.6(2) to be empty. For (2), by (1) and Proposition 4.6(1), it is enough to prove that the empty sieve covers X for the canonical topology; equivalently, by 2.6, it is enough to prove that every object $Y \to X$ over X is initial, which follows from the definition of strict initial object. □

Corollary 4.6.2. Let C be a $\mathcal{U}$-site, and let $(s_i : X_i \to X)_{i \in I}$ be a family of base-changeable morphisms in C, all with target X, satisfying:

α the family s_i is covering;

β each $s_i : X_i \to X$ is a monomorphism;

γ for each pair $i \neq j$, the object $X_i \times_X X_j$ is covered by the empty sieve, i.e. for every sheaf F on C, the set $F(X_i \times_X X_j)$ is a singleton.

Then $\epsilon_C(X)$ is the sum of the $\epsilon_C(X_i)$, $i \in I$.

Proof. This follows immediately from Proposition 4.6(2) and Corollary 4.6.1(1). □

Corollary 4.6.3. Let C be a $\mathcal{U}$-category and let $(s_i : X_i \to X)_{i \in I}$ be a family of base-changeable morphisms with the same target. Suppose that the canonical topology of C is a $\mathcal{U}$-topology. The following conditions are equivalent:

(i) There exists a $\mathcal{U}$-topology $\mathcal{T}$ on C, less fine than the canonical topology, such that $\epsilon_{\mathcal{T}}(X)$ is the sum of the $\epsilon_{\mathcal{T}}(X_i)$, $i \in I$.

(ii) The object X is the disjoint universal sum, in C, of the X_i (4.5).

Proof. (ii) $\Rightarrow$ (i). Since X is the universal sum of the X_i, the family s_i is covering for the canonical topology of C (2.6). Thus condition α of Corollary 4.6.2 holds for the canonical topology. Condition β is immediate, and condition γ follows from Proposition 4.6(1),(2). Hence (i) holds.

(i) $\Rightarrow$ (ii). Let $\mathcal{T}$ be a topology on C, less fine than the canonical topology, such that $\epsilon_{\mathcal{T}}(X)$ is the sum of the $\epsilon_{\mathcal{T}}(X_i)$. For every sheaf F for $\mathcal{T}$, the canonical map

$$\mathrm{Hom}(X, F) \to \prod_i \mathrm{Hom}(X_i, F)$$

is a bijection. Since every representable presheaf is a sheaf, it follows at once that X is the sum in C of the X_i. Applying the same argument when I is empty, we see that if an object of C is transformed into the initial object of $\widetilde{C}_{\mathcal{T}}$, then it is an initial object of C. The functor $\epsilon_{\mathcal{T}} : C \to \widetilde{C}_{\mathcal{T}}$ is fully faithful and commutes with finite projective limits. Hence condition (b) of Corollary 4.6.2 implies that the s_i are monomorphisms in C, and condition (c), by what has just been proved, implies that $X_i \times_X X_j$ is an initial object of C for $i \neq j$. Thus X is a disjoint sum of the X_i. Since $\epsilon_{\mathcal{T}}$ commutes with fiber products, this property is stable under every base change $Y \to X$, and therefore X is the disjoint universal sum of the X_i. $\square$

Corollary 4.6.4. Let C be a $\mathcal{U}$-category and let X be an object of C. Suppose that the canonical topology on C is a $\mathcal{U}$-topology. The following properties are equivalent:

(i) There exists on C a topology less fine than the canonical topology such that $\epsilon_{\mathcal{T}}(X)$ is an initial object of $\widetilde{C}_{\mathcal{T}}$.

(ii) X is a strict initial object of C.

Proof. Take I to be empty in Corollary 4.6.3. $\square$

Proposition 4.7. Let C be a $\mathcal{U}$-category, let $\widetilde{C}$ be the category of sheaves on C for the canonical topology, let

$$\epsilon_C : C \to \widetilde{C}$$

be the canonical functor (4.4.0), and let $R \rightrightarrows X$ be an equivalence relation in C admitting a cokernel Y in C. Let $\pi : X \to Y$ be the canonical morphism and suppose it is base-changeable. Consider:

(i) $\epsilon_C(Y)$ is the quotient of the equivalence relation

$$\epsilon_C(R) \rightrightarrows \epsilon_C(X).$$

(ii) The equivalence relation $R \rightrightarrows X$ is effective and universal (cf. no. 7).

Then (ii) implies (i). If C has a $\mathcal{U}$-topology less fine than the canonical topology, then (i) implies (ii).

Proof. (ii) $\Rightarrow$ (i). The canonical morphism $R \to X \times_Y X$ is an isomorphism. Let $S \to Y$ be the sieve generated by $\pi : X \to Y$. For every presheaf F, the diagram

$$\mathrm{Hom}(S, F) \longrightarrow \mathrm{Hom}(X, F) \rightrightarrows \mathrm{Hom}(R, F)$$

is exact (Exposé I, 2.12). It is therefore enough to show that, for every sheaf F for the canonical topology, the map

$$\mathrm{Hom}(Y, F) \longrightarrow \mathrm{Hom}(S, F)$$

is a bijection; equivalently, that S is covering for the canonical topology, or that the sieve generated by $\pi : X \to Y$ is covering. This follows from Definition 2.5.

(i) $\Rightarrow$ (ii). The functor $\epsilon_C : C \to \widetilde{C}$ commutes with finite projective limits and is fully faithful. Since the canonical morphism

$$\epsilon_C(R) \to \epsilon_C(X) \times_{\epsilon_C(Y)} \epsilon_C(X)$$

is an isomorphism (4.3), the canonical morphism $R \to X \times_Y X$ is an isomorphism. The morphism $\epsilon_C(X) \to \epsilon_C(Y)$ is an epimorphism; hence, by Theorem 4.4, $\pi : X \to Y$ is a covering morphism for the canonical topology. $\square$

Proposition 4.8. Let C be a $\mathcal{U}$-site. The category of sheaves on C has the following properties:

a) finite projective limits are representable;

b) direct sums indexed by an element of $\mathcal{U}$ are representable, and are disjoint and universal (4.5);

c) equivalence relations are effective and universal.

These facts were proved in Theorem 4.1(2),(3), Proposition 4.3(3), and Example 4.5.1.

These properties have been singled out because they will later make it possible to characterize the $\mathcal{U}$-categories equivalent to categories of sheaves on categories belonging to $\mathcal{U}$ (Exposé IV, 1.2).

Remark. Recall that property (a) is equivalent to the following: there exists a final object, and fiber products are representable (Exposé I, 2.3.1).

4.9

Let C be a site whose underlying category is a $\mathcal{U}$-category. Then the category of $\mathcal{U}$-sheaves on C satisfies the conditions considered in Exposé I, 7.3, either (i) or (ii); hence, by loc. cit., the various variants considered in Exposé I, 7.1 for the notion of a *generating family* in C coincide. With this understood:

Proposition 4.10. With the preceding notation, consider the canonical functor

$$\epsilon : C \to \widetilde{C}$$

(4.4.0), and let $G \subset \mathrm{Ob}(C)$ be a topologically generating family in C (3.0.1). Then the family $(\epsilon(X))_{X\in G}$ of objects of $\widetilde{C}$ is a generating family. In particular, the family $(\epsilon(X))_{X\in \mathrm{Ob}(C)}$ is generating.

We must prove that every morphism $u : F \to F'$ in $\widetilde{C}$ such that

$$\mathrm{Hom}(\epsilon(X), F) \longrightarrow \mathrm{Hom}(\epsilon(X), F') \tag{4.10.1}$$

is a bijection for every $X \in G$, is an isomorphism. The preceding map is identified with

$$u(X) : F(X) \longrightarrow F'(X), \tag{4.10.2}$$

and it remains to show that, if this map is bijective for every $X \in G$, then it is bijective for every $X \in \mathrm{Ob}(C)$. Let $(X_i \to X)_{i\in I}$ be a covering family of X by objects of G. For every pair (i, j) in $I \times I$, consider the set of all morphisms

$$X_{ijk} \to X_i \times_X X_j$$

in $\widehat{C}$, with $X_{ijk} \in G$, where k ranges over an index set $I_{i,j}$. We obtain a morphism of exact diagrams of sets

$$\begin{array}{ccccc} F(X) & \to & \prod_i F(X_i) & \rightrightarrows & \prod_{i,j,k} F(X_{ijk}) \\ \downarrow & & \downarrow\wr & & \downarrow\wr \\ F'(X) & \to & \prod_i F'(X_i) & \rightrightarrows & \prod_{i,j,k} F'(X_{ijk}), \end{array}$$

where the last two vertical arrows are bijections by hypothesis. Hence the first vertical arrow is a bijection as well. ☐

Corollary 4.11. Suppose that C is a $\mathcal{U}$-site (3.0.2). Then $\widetilde{C}$ is a $\mathcal{U}$-category and has a $\mathcal{U}$-small generating family.

Indeed, by hypothesis one may take in Proposition 4.10 a small generating family G, which proves the existence of a small generating family. On the other hand, if $F, F' \in \mathrm{Ob}(\widetilde{C})$, a homomorphism $F \to F'$ is known once the maps (4.10.1), or equivalently (4.10.2), are known for every $X \in G$ (Exposé I, 7.1.1). Hence the map

$$\mathrm{Hom}(F, F') \longrightarrow \prod_{X \in G} \mathrm{Hom}(F(X), F'(X))$$

is injective. Since the right-hand side is $\mathcal{U}$-small, the left-hand side is $\mathcal{U}$-small, which proves that $\widetilde{C}$ is a $\mathcal{U}$-category.

Corollary 4.12. Let C be a $\mathcal{U}$-site. Then for every object X of $\widetilde{C}$, both the set of subobjects of X and the set of quotient objects of X are $\mathcal{U}$-small.

This follows from Exposé I, 7.4 and 7.5, respectively, which apply by Corollary 4.11.

5 Extension of a topology from C to $\widehat{C}$

Proposition 5.1. Let C be a $\mathcal{U}$-site and let $f : H \to K$ be a morphism of $\widehat{C}$. The following conditions are equivalent:

(i) For every $X \to K$, with $X \in \mathrm{Ob}(C)$, the corresponding morphism

$$H \times_K X \to X$$

has as image a covering sieve of X.

(ii) The morphism of associated sheaves

$$\underline{a}(f) : \underline{a}H \to \underline{a}K$$

is an epimorphism in $\widetilde{C}$.

(ii bis) For every sheaf F on C, the map

$$\mathrm{Hom}(K, F) \to \mathrm{Hom}(H, F)$$

deduced from f is injective.

Proof. It is clear that (ii) is equivalent to (ii bis).

We prove (i) ⇒ (ii). By Theorem 4.4, (i) means that

$$\underline{a}(H \times_K X) \to \underline{a}(X)$$

is an epimorphism. Since

$$\underline{a}(H \times_K X) \simeq \underline{a}(H) \times_{\underline{a}(K)} \underline{a}(X),$$

because $\underline{a}$ commutes with finite projective limits (4.1), the morphism in question is obtained from

$$\underline{a}(f) : \underline{a}(H) \to \underline{a}(K)$$

by base change. As epimorphisms in $\widetilde{C}$ are universal, this shows that (ii) implies (i) after base change; conversely, since the family of all $X \to K$ is epimorphic and the corresponding family $\underline{a}(X) \to \underline{a}(K)$ is epimorphic in $\widetilde{C}$ (4.1), to verify that $\underline{a}(f)$ is an epimorphism it is enough to check it after every such base change. This proves (i) $\Rightarrow$ (ii), and the equivalence follows. $\square$

Definition 5.2.

1) A morphism $f : H \to K$ satisfying the equivalent conditions of Proposition 5.1 is called a *covering morphism*. A family of morphisms $(f_i : H_i \to K)_{i \in I}$ with common target is called *covering* if the corresponding morphism

$$f : \coprod_i H_i \to K$$

is covering.

2) A morphism $f : H \to K$ is called *bicovering* if it is covering and if its diagonal morphism

$$H \to H \times_K H$$

is covering. A family $(f_i : H_i \to K)_{i \in I}$ with common target is called *bicovering* if the corresponding morphism $\coprod_i H_i \to K$ is bicovering.

By condition (i) of Proposition 5.1, to say that a family $(f_i : H_i \to K)_{i \in I}$ is covering means that, for every morphism $X \to K$, with $X \in \mathrm{Ob}(C)$, the family

$$H_i \times_K X \to X, \qquad i \in I,$$

has as image a covering sieve of X; or equivalently, by condition (ii), that the family of morphisms

$$\underline{a}(f_i) : \underline{a}H_i \to \underline{a}K$$

in $\widetilde{C}$ is epimorphic, taking into account that $\underline{a}$ commutes with direct sums (4.1).[2]

Proposition 5.3. Let C be a $\mathcal{U}$-site and let $f : H \to K$ be a morphism of $\widehat{C}$. The following conditions are equivalent:

(i) f is bicovering (5.2).

[2]The property of being covering (respectively bicovering), for a morphism or family of morphisms of $\widehat{C}$, is stable under base change.

(i bis) f is covering, and for every object X of C and every pair of morphisms $u, v : X \rightrightarrows H$ such that $fu = fv$, the equalizer of (u, v) is a covering sieve of X.

(ii) $\underline{a}(f) : \underline{a}H \to \underline{a}K$ is an isomorphism in $\widetilde{C}$.

(ii bis) For every sheaf F on C, the map

$$\mathrm{Hom}(K, F) \to \mathrm{Hom}(H, F)$$

is a bijection.

Proof. The equivalence (i) $\Leftrightarrow$ (i bis) follows from condition (i) of Proposition 5.1, applied to the diagonal morphism $H \to H \times_K H$. The equivalence (ii) $\Leftrightarrow$ (ii bis) is trivial.

(i) $\Rightarrow$ (ii). The morphism

$$\underline{a}(f) : \underline{a}H \to \underline{a}K$$

is an epimorphism by Proposition 5.1. Since $\underline{a}$ commutes with fiber products (4.1), the diagonal

$$\underline{a}H \to \underline{a}H \times_{\underline{a}K} \underline{a}H$$

is an epimorphism by Proposition 5.1. A diagonal morphism is always a monomorphism; hence it is an isomorphism by Proposition 4.2. Thus $\underline{a}(f)$ is a monomorphism. It is therefore an isomorphism by Proposition 4.2.

(ii) $\Rightarrow$ (i). Since $\underline{a}$ commutes with fiber products, both f and the diagonal $H \to H \times_K H$ are transformed by $\underline{a}$ into isomorphisms. In particular they are transformed into epimorphisms. Hence they are covering. ☐

5.3.1

It follows from Proposition 5.3 and from the fact that $\underline{a}$ commutes with direct sums that a family

$$(f_i : H_i \to K)_{i \in I}$$

of morphisms of $\widehat{C}$ is bicovering if and only if the f_i induce, on associated sheaves, an isomorphism

$$\coprod_i \underline{a}H_i \xrightarrow{\sim} \underline{a}K,$$

or equivalently if and only if, for every sheaf F, the map

$$\mathrm{Hom}(K, F) \longrightarrow \prod_i \mathrm{Hom}(H_i, F)$$

is bijective.

Proposition 5.4. Let C be a $\mathcal{U}$-site. There exists on $\widehat{C}$ a topology, clearly unique, such that a family $H_i \to K$ of arrows of $\widehat{C}$ with common target is covering for this topology if and only if it is covering in the sense of Definition 5.2. It is also the least fine topology on $\widehat{C}$ among the topologies T satisfying:

a) T is finer than the canonical topology of $\widehat{C}$, i.e. every epimorphic family in $\widehat{C}$ is covering for T;

b) every covering family in C is covering in $\widehat{C}$.

Proof. We only give indications. It is left to the reader to show that the families covering in the sense of Definition 5.2 are the covering families of a topology T_C on $\widehat{C}$; one uses Theorem 4.1. The covering families for the canonical topology on $\widehat{C}$ are the epimorphic families of $\widehat{C}$ (2.6 and Exposé I, 3.1). Since $\underline{a}$ commutes with inductive limits (4.1), the topology T_C is finer than the canonical topology of $\widehat{C}$. Moreover, the covering families of C are covering families for T_C (5.1).

Let T' denote the least fine topology on $\widehat{C}$ having properties (a) and (b). Then T_C is finer than T'. Let

$$(f_i : H_i \to K)_{i \in I}$$

be a T_C-covering family. We show that it is a covering family for T'. Let $s_i : H_i \to H = \coprod_i H_i$ be the canonical monomorphisms, and let $f = (f_i) : H \to K$ be the morphism defined by the f_i. The family s_i is covering for T'. Thus, by axiom (T2), to prove that (f_i) is covering for T', it is enough to prove that the morphism $f : H \to K$ is covering for T'.

There exists an epimorphic family in $\widehat{C}$

$$u_\lambda : X_\lambda \to K, \qquad \lambda \in \Lambda, \quad X_\lambda \in \mathrm{Ob}(C)$$

(Exposé I, 3.4). By axiom (T2), to prove that $f : H \to K$ is covering for T', it is enough to prove that, for every λ, the projection

$$H \times_K X_\lambda \to X_\lambda$$

is covering for T'. Let

$$v_j : Y_j \to H \times_K X_\lambda, \qquad j \in J, \quad Y_j \in \mathrm{Ob}(C),$$

be an epimorphic family in $\widehat{C}$. The family $(\mathrm{pr}_2 \circ v_j)_{j \in J}$ is covering for T_C. Hence it is a covering family in C by Proposition 5.1, and therefore is covering for T'. Consequently the sieve generated by $H \times_K X_\lambda \to X_\lambda$ contains a T'-covering sieve; it is therefore covering for T'. This proves the proposition. $\square$

Remark 5.4.1. The proof of Proposition 5.4 actually shows that every topology T' on $\widehat{C}$ which is finer than the canonical topology of $\widehat{C}$, is the least fine of the topologies T on $\widehat{C}$ satisfying:

a) T is finer than the canonical topology of $\widehat{C}$;

b) every T'-covering family of the form

$$u_i : X_i \to X, \qquad i \in I,$$

where X and the X_i are objects of C, is covering for T.

In particular, every topology T' on $\widehat{C}$ finer than the canonical topology is uniquely determined by the families $u_i : X_i \to X$, $i \in I$, with X_i and X objects of C, which are covering for T'.

Remark 5.4.2. One can easily show that, for every topology T' on $\widehat{C}$ finer than the canonical topology, the set of families of morphisms

$$(X_i \to X)_{i\in I}$$

with common target in C, which are covering for T', is the set of covering families of a topology on C. Thus Proposition 5.4 and Remark 5.4.1 establish a one-to-one correspondence between topologies on C and topologies on $\widehat{C}$ finer than the canonical topology.

5.5.0

Let C be a small category. Let $\mathcal{C}$ be the set of strictly full subcategories of $\widehat{C}$ (strictly full means that every object isomorphic to an object of the subcategory is itself an object of the subcategory) whose inclusion functor admits a left adjoint commuting with finite projective limits. Also let $\mathcal{T}$ denote the set of topologies on C. Theorem 3.4 defines a map

$$\Phi : \mathcal{T} \longrightarrow \mathcal{C}.$$

We now define a map in the other direction. For this, to every element

$$i' : C' \hookrightarrow \widehat{C}, \qquad \underline{a}' \dashv i',$$

with $\underline{a}'$ left adjoint to i', we associate a topology T_e on C. For every object X of C, $J_e(X)$ is defined to be the set of subobjects of X whose inclusion morphism is transformed by $\underline{a}'$ into an isomorphism. Using the hypotheses on $\underline{a}'$, one immediately checks that this defines a topology T_e on C. We have therefore defined a map

$$\Psi : \mathcal{C} \longrightarrow \mathcal{T}.$$

We then have the following result, due to J. Giraud.

Theorem 5.5. The map Φ is a bijection, and Ψ is its inverse.

Proof. The composite $\Psi \circ \Phi$ is the identity. Indeed, this follows immediately from Proposition 5.1.

The composite $\Phi \circ \Psi$ is the identity. Let

$$i' : C' \hookrightarrow \widehat{C}, \qquad \underline{a}' \dashv i',$$

be an element of $\mathcal{C}$, let T_e be the topology corresponding to it under Ψ, and let C_e be the category of sheaves for T_e. Using the definition of the topology T_e and the definition of bicovering morphisms (5.2), one proves the equivalence of the following assertions:

(i) a morphism u of $\widehat{C}$ is bicovering for the topology T_e;

(ii) the morphism u of $\widehat{C}$ is transformed by $\underline{a}'$ into an isomorphism.

It is clear that C' is a full subcategory of C_e. It remains only to prove that every sheaf F for the topology T_e is an object of C'. By the equivalence above, the morphism

$$F \longrightarrow i'\underline{a}'(F)$$

is bicovering; since its source and target are sheaves, Proposition 5.3(ii) implies that it is an isomorphism. $\square$

6 Sheaves with values in a category

6.0

Let C and D be two categories. A contravariant functor

$$F : C^\circ \to D$$

is called a *presheaf on C with values in D*.

Definition 6.1. Let C be a site and let D be a category. A presheaf $F : C^\circ \to D$ is called a *sheaf on C with values in D*, or more briefly a *D-valued sheaf*, if for every object S of D, the presheaf of sets

$$X \longmapsto \mathrm{Hom}_D(S, F(X)), \qquad X \in \mathrm{Ob}(C),$$

is a sheaf. The full subcategory of $\mathrm{Hom}(C^\circ, D)$ consisting of sheaves on C with values in D will be denoted

$$\mathrm{Hom}^\sim(C^\circ, D).$$

Remarks 6.2.

1) Condition 6.1 means that for every object X of C, every covering sieve R of X, and every object S of D, one has an isomorphism

$$\mathrm{Hom}_D(S, F(X)) \xrightarrow{\sim} \varprojlim_{Y \to R} \mathrm{Hom}_D(S, F(Y))$$

(Exposé I, 3.5 and 2.1). When D is the category of $\mathcal{U}$-sets, this recovers the definition of sheaves of sets.

2) Let D' be a $\mathcal{U}$-category and let $G : D \to D'$ be a functor commuting with projective limits. By composition, G sends sheaves with values in D to sheaves with values in D'.

6.3.0

Let γ be a species of algebraic structure defined by finite projective limits (Exposé I, 2.9), and let $\mathcal{U}$-γ be the category of γ-sets which belong to $\mathcal{U}$. Denote by

$$\mathrm{For} : \mathcal{U}\text{-}\gamma \to \mathcal{U}\text{-}\mathbf{Set}$$

the forgetful functor to the underlying set. The functor For commutes with projective limits and therefore, by the preceding remark, defines by composition functors

$$\mathrm{For}^{\widehat{\ }} : \mathrm{Hom}(C^\circ, \mathcal{U}\text{-}\gamma) \longrightarrow \widehat{C},$$

and

$$\mathrm{For}^{\widetilde{\ }} : \mathrm{Hom}^{\sim}(C^\circ, \mathcal{U}\text{-}\gamma) \longrightarrow \widetilde{C}.$$

The latter is called the *underlying sheaf of sets* functor. In fact, it factors canonically through the category γ-$\widetilde{C}$ of γ-objects of $\widetilde{C}$. Denoting the resulting functor again by

$$\widetilde{\mathrm{For}} : \mathrm{Hom}^{\sim}(C^\circ, \mathcal{U}\text{-}\gamma) \longrightarrow \gamma\text{-}\widetilde{C},$$

one may state:

Proposition 6.3.1. The functor $\mathrm{For}^{\widetilde{\ }}$ establishes equivalences of categories between:

1) sheaves on C with values in $\mathcal{U}$-γ;

2) γ-objects of the category of sheaves of sets;

3) γ-objects of the category of presheaves of sets whose underlying presheaf is a sheaf.

Proof. The proof uses essentially Exposé I, 3.2 and the fact that a finite projective limit of sheaves, computed in the category of presheaves, is again a sheaf (4.1). □

6.3.2

The preceding proposition justifies the abuse of language by which one identifies a sheaf with values in $\mathcal{U}$-γ with the corresponding γ-object in $\widetilde{C}$. We shall henceforth use this abuse of language systematically.

We shall use the classical terminology: *sheaves of groups*, *sheaves of commutative groups*, which we shall most often call *abelian sheaves*, *sheaves of rings*, *sheaves of A-modules*, and so on.

6.3.3

We denote by

$$\widetilde{C}_\gamma \quad (\text{respectively } \widehat{C}_\gamma)$$

the category of γ-objects of $\widetilde{C}$ (respectively of $\widehat{C}$). When γ is the species of structure "*A*-module", we also write $\widetilde{C}_A$, or simply $\widetilde{C}_{\mathrm{ab}}$ when $A = \mathbb{Z}$.

Assume that C is a $\mathcal{U}$-site. The associated-sheaf functor is left exact (4.1), and therefore:

Proposition 6.4. The inclusion functor

$$i_\gamma : \widetilde{C}_\gamma \to \widehat{C}_\gamma$$

has a left adjoint

$$a_\gamma : \widehat{C}_\gamma \to \widetilde{C}_\gamma$$

which is left exact; in other words, there is an isomorphism

$$\mathrm{Hom}_{\widetilde{C}_\gamma}(a_\gamma(\,\cdot\,),\,\cdot\,) \xrightarrow{\sim} \mathrm{Hom}_{\widehat{C}_\gamma}(\,\cdot\,, i_\gamma(\,\cdot\,)).$$

Let X be an object of $\widehat{C}_\gamma$. The underlying sheaf of sets of $a_\gamma(X)$ is canonically isomorphic to the sheaf of sets associated to the underlying presheaf of sets of X. The morphism of underlying presheaves of sets associated to the adjunction morphism

$$\mathrm{id} \to i_\gamma a_\gamma$$

is identified with the adjunction morphism applied to the underlying presheaf of sets.

Proposition 6.5. Suppose that the forgetful functor

$$\mathrm{For} : \gamma\text{-}\mathcal{U} \to \mathcal{U}\text{-}\mathbf{Set}$$

has a left adjoint[3]

$$\mathrm{Lib} : \mathcal{U}\text{-}\mathbf{Set} \to \gamma\text{-}\mathcal{U},$$

i.e.

$$\mathrm{Hom}_{\mathcal{U}\text{-}\mathbf{Set}}(\,\cdot\,, \mathrm{For}(\,\cdot\,)) \xrightarrow{\sim} \mathrm{Hom}_{\gamma\text{-}\mathcal{U}}(\mathrm{Lib}(\,\cdot\,),\,\cdot\,).$$

Then the functor

$$\mathrm{For}^{\sim} : \widetilde{C}_\gamma \to \widetilde{C}$$

(respectively

$$\mathrm{For}^{\wedge} : \widehat{C}_\gamma \to \widehat{C})$$

has a left adjoint

$$\mathrm{Lib}^{\sim} \quad (\text{respectively } \mathrm{Lib}^{\wedge}),$$

and there is a canonical isomorphism

$$\mathrm{Lib}^{\sim} = a_\gamma \,\mathrm{Lib}^{\wedge}\, i.$$

Proof. The proof is formal and is left to the reader. □

Corollary 6.6. Let $(X_i)_{i\in I}$ be a generating family of $\widetilde{C}$ (4.9). Then the family

$$\mathrm{Lib}^{\sim}(X_i)$$

[3] One proves that such a left adjoint always exists in the usual examples: free group, free abelian group, free A-module, and so on.

is a generating family of $\widetilde{C}_\gamma$.

Proof. The proof is formal once one has observed that the functor $\mathrm{For}^{\sim}$ commutes with projective limits and is conservative: $\mathrm{For}^{\sim}(u)$ is an isomorphism if and only if u is an isomorphism.
□

Proposition 6.7. Let A be a $\mathcal{U}$-sheaf of rings, or a small ring. Let

$$\widetilde{C}_A \quad \text{(respectively } \widehat{C}_A\text{)}$$

be the category of sheaves (respectively presheaves) of unital A-modules (6.3.3) on a $\mathcal{U}$-site C. Then $\widetilde{C}_A$ is a $\mathcal{U}$-abelian category satisfying the axioms (AB5), "existence of left exact filtered inductive limits", and (AB3)*, "existence of infinite products", of Tohoku. It has a family of generators indexed by an element of $\mathcal{U}$.

Proof. It is clear that $\widehat{C}_A$ is an abelian category satisfying axioms (AB3), (AB4), and (AB5) (Exposé I, 2.8 and 3.3). From Proposition 6.4 it follows that $\widetilde{C}_A$ is an additive category in which kernels and cokernels are representable.

More explicitly, let i_A and a_A be the inclusion and associated-sheaf functors, and let $u : X \to Y$ be a morphism of $\widetilde{C}_A$. The functor a_A is left exact and commutes with inductive limits. The functor i_A commutes with projective limits. Hence the functor

$$i_A a_A : \widehat{C}_A \to \widehat{C}_A$$

is additive and left exact. We get canonical isomorphisms

$$i_A(\mathrm{Ker}(u)) \xrightarrow{\sim} \mathrm{Ker}(i_A(u)), \tag{*}$$

and

$$\mathrm{coker}(u) \xrightarrow{\sim} a_A\ \mathrm{coker}(i_A(u)). \tag{**}$$

It follows that there is a canonical isomorphism

$$\mathrm{coim}(u) \xrightarrow{\sim} a_A\ \mathrm{coim}(i_A(u)). \tag{***}$$

We prove that the canonical morphism $\mathrm{coim}(u) \to \mathrm{im}(u)$ is an isomorphism. By $(*)$, it is enough to prove exactness of the sequence

$$0 \longrightarrow i_A(\mathrm{coim}(u)) \longrightarrow i_A(Y) \longrightarrow i_A(\mathrm{coker}(u)).$$

Using the isomorphisms $(**)$ and $(***)$, and observing that $a_A i_A(Y)$ is isomorphic to Y, this sequence is identified with

$$0 \longrightarrow i_A a_A(\mathrm{coim}(i_A(u))) \longrightarrow i_A a_A i_A(Y) \longrightarrow i_A a_A(\mathrm{coker}(i_A(u))),$$

which is simply the image under the left exact functor $i_A a_A$ of the exact sequence

$$0 \longrightarrow \mathrm{coim}(i_A(u)) \longrightarrow i_A(Y) \longrightarrow \mathrm{coker}(i_A(u)).$$

The category $\widetilde{C}_A$ is therefore abelian.

The functor

$$a_A : \widehat{C}_A \to \widetilde{C}_A$$

is left exact and commutes with arbitrary inductive limits. Hence it is additive, exact, and commutes with inductive limits. Since $\widehat{C}_A$ satisfies axiom (AB5), so does $\widetilde{C}_A$. Finally, it is clear that in $\widetilde{C}_A$, products indexed by an element of $\mathcal{U}$ are representable; hence axiom (AB3)* holds. This completes the proof of the first assertion.

For every ring A which is an element of $\mathcal{U}$, the functor

$$\mathrm{For} : \mathcal{U}\text{-}A\text{-Mod} \longrightarrow \mathcal{U}\text{-}\mathbf{Set}$$

has a left adjoint X_A, the free A-module generated by X. It remains only to apply Proposition 4.10 and Corollary 6.6. □

Notation 6.8. Let H be a sheaf of sets. The sheaf of A-modules associated to the presheaf (cf. notation 6.5)

$$X \longmapsto \mathrm{Lib}_A\, H(X), \qquad X \in \mathrm{Ob}(C),$$

is denoted A_H. When $H = \underline{a}(X)$, the sheaf associated to the presheaf represented by X, one sometimes simply writes A_X, by abuse of notation. The proof of Proposition 6.7 shows that the family of sheaves of A-modules A_X, $X \in \mathrm{Ob}(C)$, is a generating family, indexed by an element of $\mathcal{U}$, for the category of sheaves of A-modules.

Remark 6.9. By Tohoku, Proposition 6.7 shows that when A is an element of $\mathcal{U}$, the category $\widetilde{C}_A$ has enough injectives. It is known, moreover, that infinite products are not necessarily exact in $\widetilde{C}_A$, and therefore that the category $\widetilde{C}_A$ does not in general have enough projectives [Roos].

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